

ME701 Embedded Systems 1 Introduction

Dr John Wade
Email: john.wade@atu.ie



Module Overview

Teaching and Assessment Methods

Module Delivery



- **Teaching methods:**
 - Lectures:~ 2hrs per week
 - Mondays 9:30 – 11:30 in W1115
 - Lab Practicals:~ 4hrs per week
 - Tuesdays 11.30 – 1:30 in W1128
 - Thursdays 1.30 – 3.30 in W1128
- **Module website (*Blackboard*)**
 - All Learning and Teaching Materials Available via Blackboard
 - All Assignment Submissions through BBL

Module Assessment

40% Continuous Assessment (CA)

- **Coursework (40%):**
 - Part A Complete Mini Project 1 (20%)
 - Series of Labs and a mini Microcontroller based project using MSP430 Launchpad which utilises ADC, LCD Screen & servo/stepper
 - Part B Mini Project 2 (20%)
 - Series of FPGA Labs.

Module Assessment

60% End Exam

- **Exam (60%):**
 - .3 Hour Duration
 - 2 Sections:
 - A. Microcontrollers
 - B. VHDL/FPGAs
 - 7 Questions in total: Answer 5 with at least 2 from section A and two from section B

What are Embedded Systems??

Embedded Systems

What Are They??

- An embedded system is one that has computer hardware with software embedded in it as one of its components

Or

- We can define an embedded system as “A microprocessor based system that does not look like a computer”.

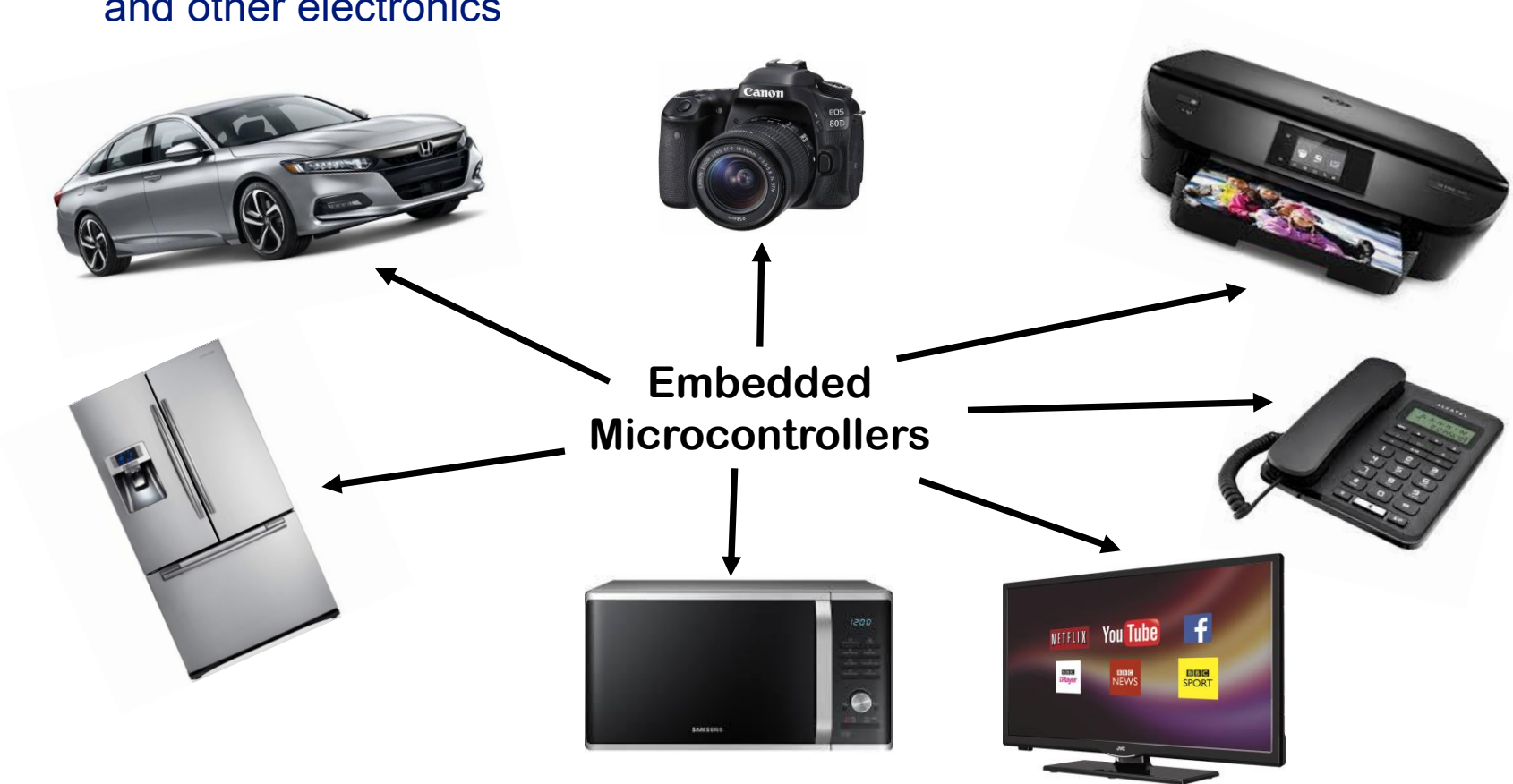
Or

- we can say that it is “A combination of computer hardware and software, and perhaps additional mechanical or other parts, designed to perform a dedicated function. In some cases, embedded systems are part of a larger system or product, as is the case of an antilock braking system in a car ”.

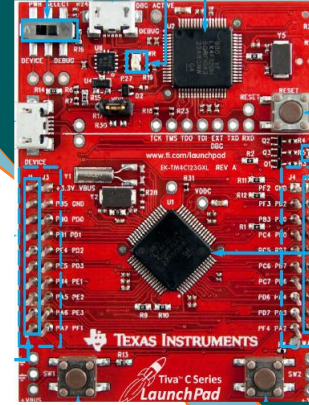
Introduction to Microcontrollers

Where Do You Find Them?

- Microcontrollers are hidden in a huge number of appliances, gadgets, and other electronics



Microcontrollers



- **Microcontroller Unit (MCU):**
 - When used to control electronic device its called an (Embedded System)
 - By combining microprocessors with memory units and Input/Output (I/O) ports, a Microcontroller system can be built.
 - Made by embedding microprocessors, memory unit and I/O ports into a single semiconductor chip.
 - Example: MSP430 (used in this module).
 - You should already be familiar with how to do some basic GPIO programing of these boards in C using Code Composer Studio.

Analog to Digital Convertors (ADC)

Overview

- Introduction to Analog - MCU Interfacing
 - Why its needed
- Analog to Digital Convertors
 - Quantization
 - Encoding
 - Errors
 - Sampling
 - Types of ADC
- TIVA C Launchpad ADC Programming
 - Registers



Introduction

Why Analog interfacing is Needed

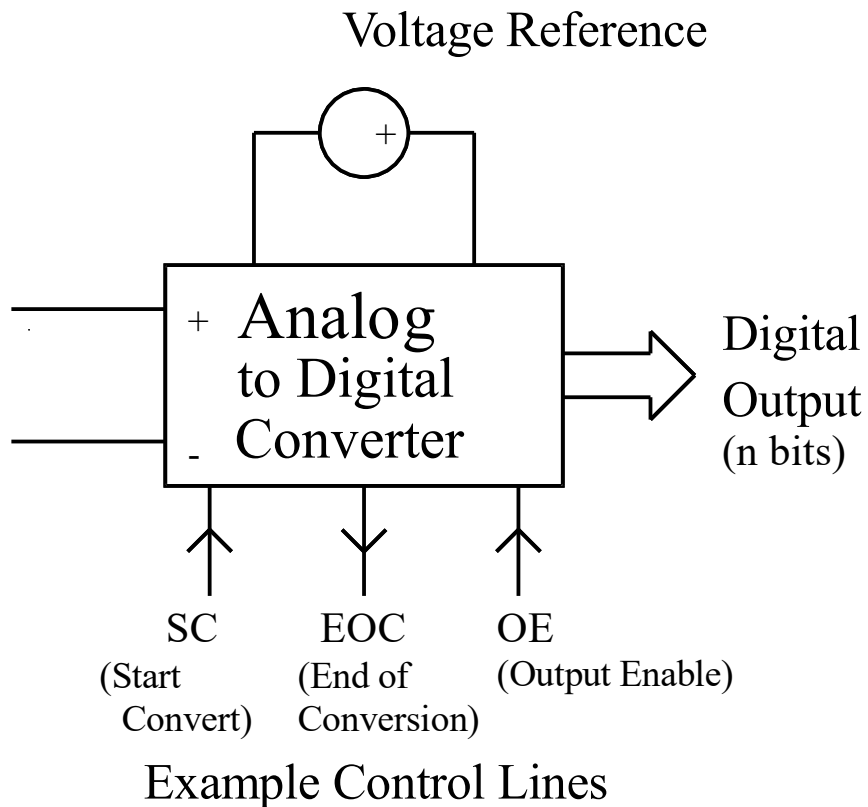
- Embedded systems often need to measure values of physical parameters from the real world
- These parameters are usually continuous (*analog*) and not in a digital form which computers (which operate on discrete data values) can process.
 - This type of system is called a Mixed Signal System as it uses a mixture of analogue and Digital signals. Analogue from the real world and Digital within the embedded system
- **Some Examples:**
 - **Temperature**
 - Thermometer (do you have a fever?)
 - Thermostat for building, fridge, freezer
 - Car engine controller
 - Chemical reaction monitor
 - Safety (e.g. microprocessor processor thermal management)
 - **Light (or infrared or ultraviolet) intensity**
 - Digital camera
 - IR remote control receiver
 - Tanning bed
 - UV monitor
 - **Rotary position**
 - Wind gauge
 - Knobs
 - **Pressure**
 - Blood pressure monitor
 - Altimeter
 - Car engine controller
 - Scuba dive computer
 - Tsunami detector
 - **Acceleration**
 - Air bag controller
 - Vehicle stability
 - Video game remote
 - **Microphones**
 - Voice Recording
 - **Other**
 - Touch screen controller
 - EKG, EEG
 - Breathalyzer
 - Many many more

Analog to Digital Convertors

ADC

- As mentioned before, most sensors of temperature, sound, and acceleration have analog outputs. However, these analog outputs have to be converted into digital signals before they can be stored or used in calculations by the microcontroller
- Analog signals are converted into digital representations with a resolution and a rate determined by the Analog to Digital Converter (ADC)
- An ADC is a circuit whose digital output is proportional to the analog input
- Effectively, an ADC measures the input analog voltage, and gives a binary output proportional to the size of the input signal
- The input range of the ADC is usually determined by the value of a voltage reference, V_{ref} (e.g. 3.3 V)

Generic ADC Block Diagram



- The conversion process is started by a digital input called **SC**, **start convert**
- It takes a finite time to perform the conversion, depending on the clock frequency
- The ADC signals that the conversion is completed via the **EOC**, **end of conversion** line
- The resulting data can then be written onto the output data bus, when the **output enable**, **OE** control line, is high

Depth Gauge Example

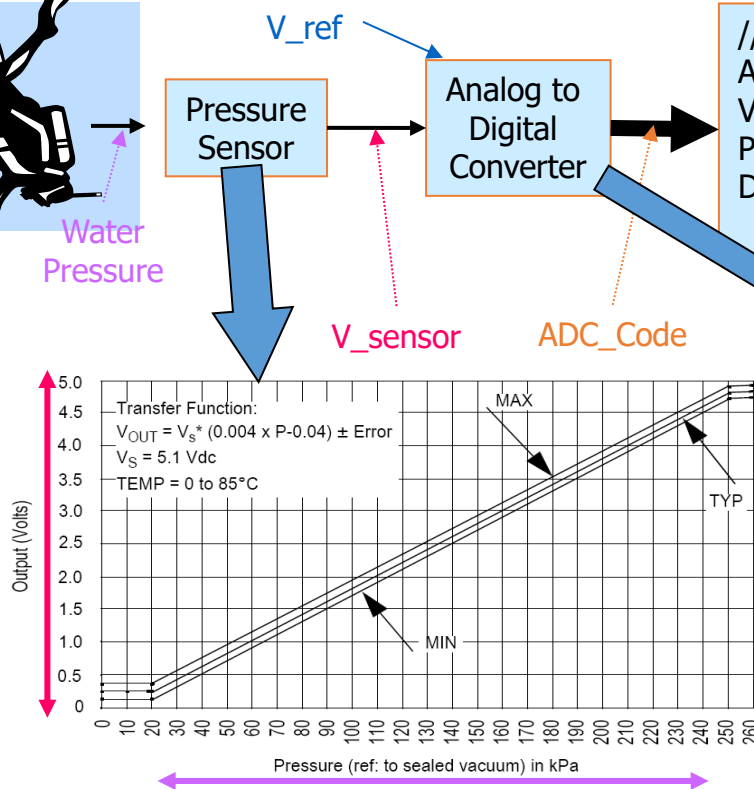


Figure 4. Output vs. Absolute Pressure

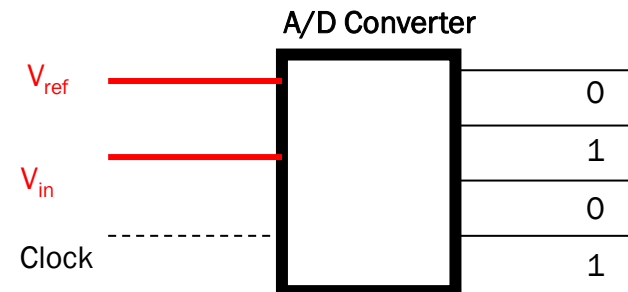
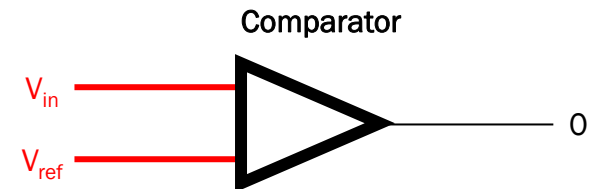
```
// Your software
ADC_Code = ADC0_R[0];
V_sensor = ADC_code*V_ref/1023;
Pressure_kPa = 250 * (V_sensor/V_supply+0.04);
Depth_ft = 33 * (Pressure_kPa -
    Atmos_Press_kPa)/101.3;
```

Voltages	ADC Output Codes
V_{ref}	111..111 111..110 111..101 111..100
V_{sensor}	ADC_Code
Ground	000..001 000..000

1. Sensor detects *water pressure* and generates a proportional *output voltage* V_{sensor}
2. ADC generates a proportional digital *integer* (code) based on V_{sensor} and V_{ref}
3. Code can convert that integer to something more useful
 1. first a float representing the *voltage*,
 2. then another float representing *pressure*,
 3. finally another float representing *depth*

Getting from Analog to Digital

- A **Comparator** tells us “Is $V_{in} > V_{ref}$?”
 - Compares an **analog input voltage** (V_{in}) with an **analog reference voltage** (V_{ref}) and determines which is larger, returning a 1-bit number
- An **Analog to Digital converter** [ADC] tells us how large V_{in} is as a fraction of V_{ref} .
 - Reads an analog input signal (usually a voltage) and produces a corresponding multi-bit number at the output.
 - E.g. calculate the depth



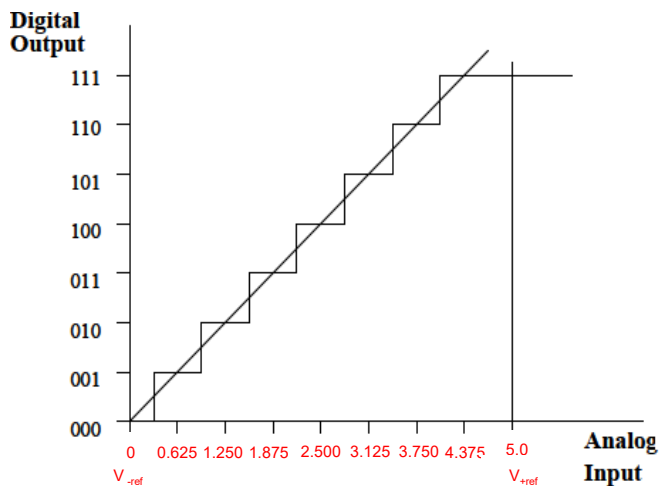
ADC -> 2 Step Process

- **Quantizing** - breaking down analog value to set of finite states
- **Encoding** - assigning a digital word or number to each state

Step 1: Quantizing

Example: a 3 bit A/D , $N=2^3=8$ (no. of steps (resolution)), $V_{+ref} = 5V$, $V_{-ref} = 0V$

0-5V (V_{-ref} to V_{+ref}) signals
Separated into discrete states
with 0.625V increments.



Analog quantization size (Q):
 $Q = (V_{+ref} - V_{-ref}) / N = (5V - 0V) / 8 =$
0.625V

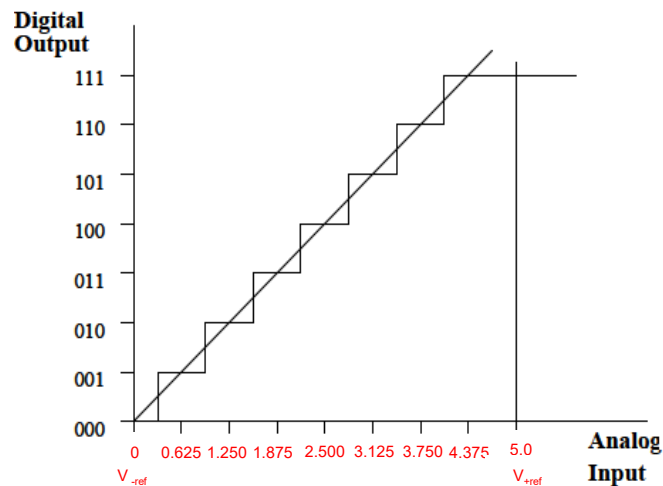
This is also known as the
resolution of the ADC (V_{res})

Output States	Discrete Voltage Ranges (V)
0	0.000 - 0.625
1	0.625 - 1.250
2	1.250 - 1.875
3	1.875 - 2.500
4	2.500 - 3.125
5	3.125 - 3.750
6	3.750 - 4.375
7	4.375 - 5.000

Step 2: Encoding

Give a value to each state

0-5V (V_{-ref} to V_{+ref}) signals
Separated into discrete states
with 0.625V increments.



Analog quantization size (Q):
 $Q = (V_{+ref} - V_{-ref}) / N = (5V - 0V) / 8 =$
0.625V

Output States	Output Binary Code
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Forward Transfer Function Equations

Calculating output of ADC

What code N_{out} will the ADC use to represent voltage V_{in} ?

General Equation

$$N_{out} = \frac{(V_{in} - V_{-ref}) \times (2^n - 1)}{V_{REF}}$$

Simplification with $V_{-ref} = 0 \text{ V}$

$$N_{out} = \frac{V_{in} \times (2^n - 1)}{V_{REF}}$$

N_{out} = converted code

V_{in} = sampled input voltage

V_{+ref} = upper voltage reference

V_{-ref} = lower voltage reference

N = number of bits of resolution in ADC

Note:

N_{out} is always rounded to the nearest integer value

Example: An 8-bit ADC has a reference voltage of 5V. Calculate the output number N_{out} if the $V_{in} = 1\text{V}$

$$\text{Ans: } N_{out} = \frac{V_i \times (2^n - 1)}{V_{REF}} = \frac{1\text{V} \times 255}{5\text{V}} = \mathbf{51 = 0011\ 0011}$$

Inverse Transfer Function Equations

Calculating the Voltage input, V_i

What range of voltages V_{in_min} to V_{in_max} does code n represent?

General Equation

N_{out} = converted code

V_{+ref} = upper voltage reference

V_{-ref} = lower voltage reference

N = number of bits of resolution in ADC

$$V_{in} = \frac{N_{out}}{2^N - 1} (V_{+ref} - V_{-ref})$$

Simplification with $V_{-ref} = 0\text{ V}$

$$V_{in_min} = \frac{N_{out}}{2^N - 1} (V_{+ref})$$

Example: What voltage does a reading of 676 represent for a 10 bit ADC with $V_{+ref} = 5\text{V}$?

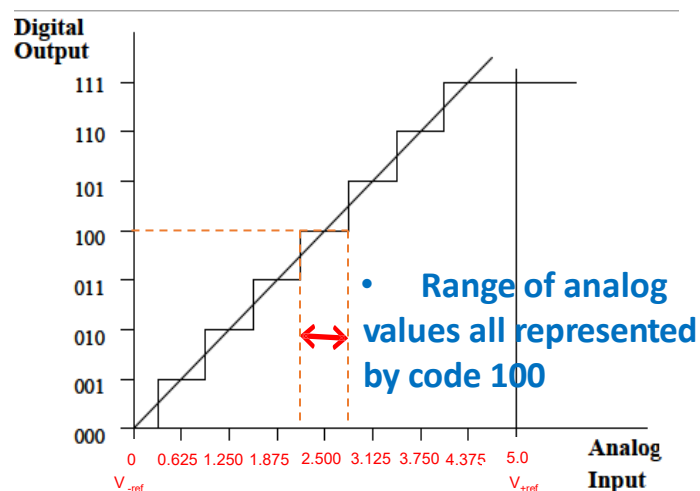
$$\text{Ans: } V_{in} = \frac{676}{2^{10} - 1} (5\text{V})$$

$$\therefore V_{in} = \frac{676}{1023} (5\text{V}) = 3.304\text{V}$$

ADC Accuracy (Error)

Quantization Error

- The greatest quantization error is one half of the step width, or half of one least significant bit (LSB) equivalent on the voltage scale:
 - Error = $0.5 * Q$**
- To reduce the quantization error, the “step width” should be narrowed. This can be achieved by increasing the number of bits (resolution) in the ADC process
 - This increases complexity and cost, and often the time taken to undertake the conversion



- As sensor voltage goes from 0 V to V_{max} , (V_{max} should not be greater than V_{+ref}) the digital output of ADC will step from 0 to 111 or (2^3-1)
- $Q = V_{max} / (2^3) = 0.625$
- Example:** Error for code 100 = $2.5V \pm 0.3125V$

Resolution versus Step Size for ADC

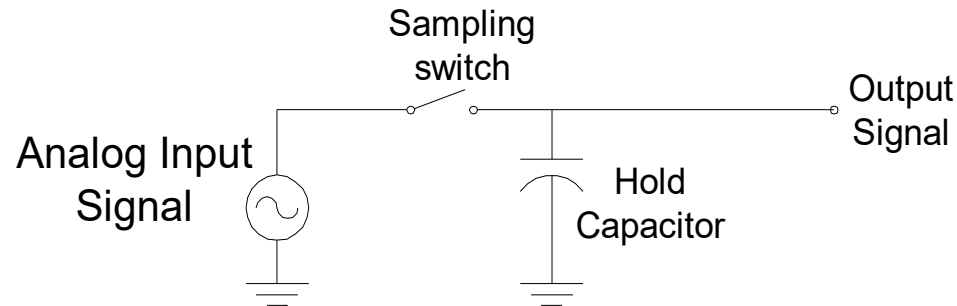
$$V_{+ref} = 5V$$

n-bit	Number of steps	Step size
8	256	$5V / 256 = 19.53 \text{ mV}$
10	1024	$5V / 1024 = 4.88 \text{ mV}$
12	4096	$5V / 4096 = 1.2 \text{ mV}$
16	65,536	$5V / 65,536 = 0.076 \text{ mV}$
Note: $V_{ref} = 5V$		

- As the resolution of the ADC increases the voltage between ADC states decreases and therefore the accuracy increases

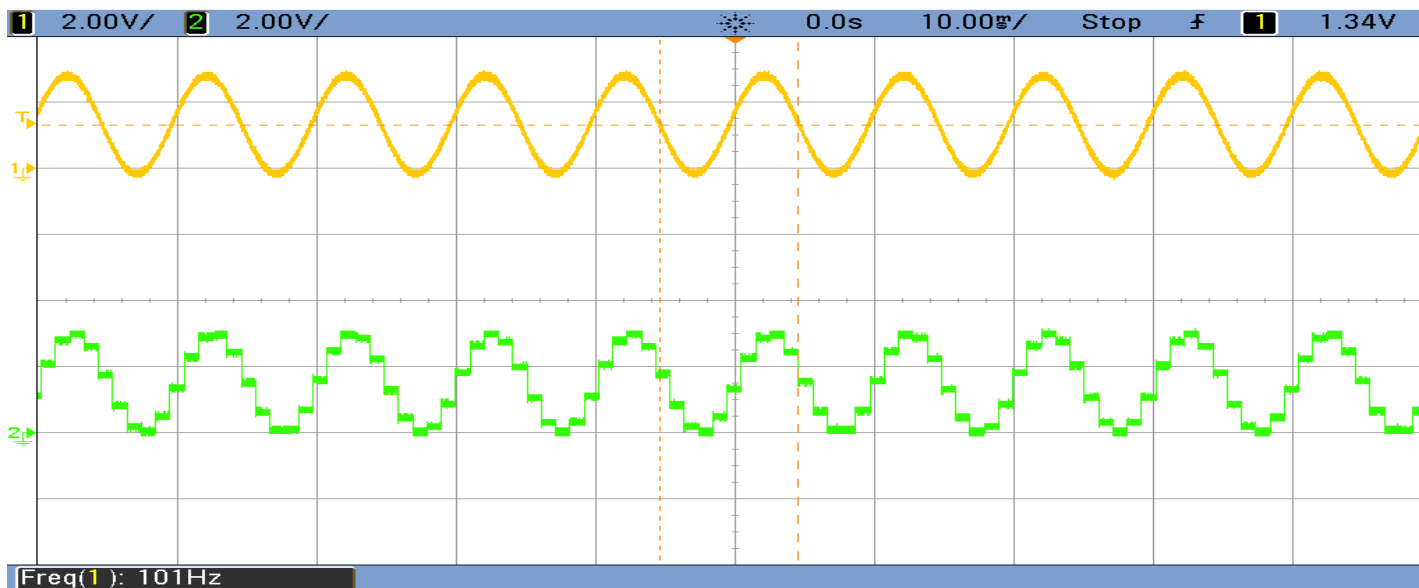
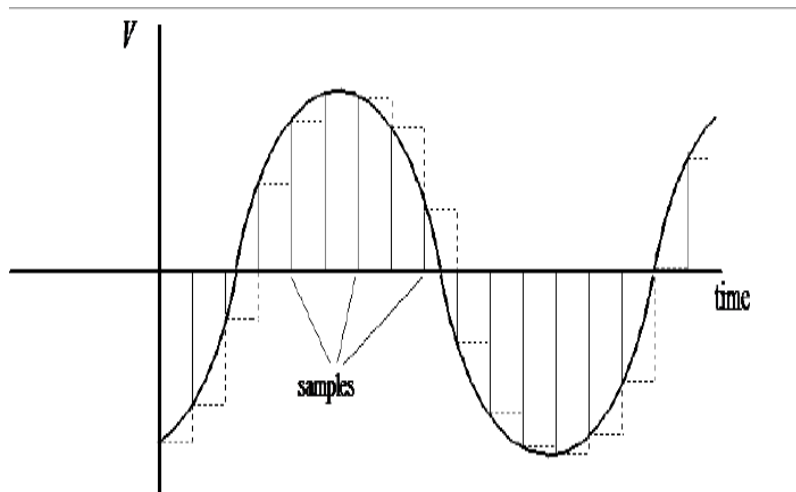
Sample and Hold

- When performing an analog to digital conversion, a “sample” is taken of analog signal and quantized to the accuracy defined by the resolution of the ADC.



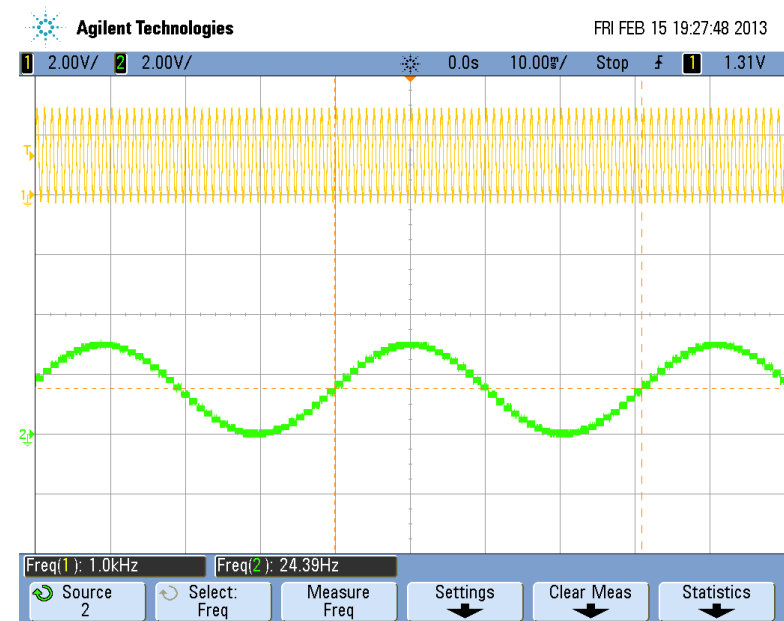
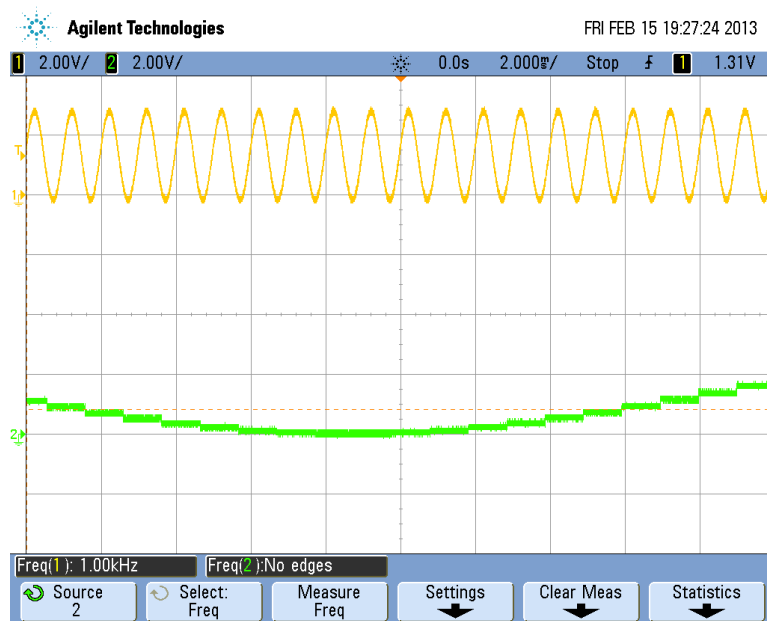
- The signal is held, which means that it is stored temporally while the signal is compared to the reference voltage and the digital signal is generated.**
- Sometimes, the digital signal is generated by averaging the results of several analog samples. The more samples taken, the more accurate the digital data will be as long as the analog signal is not changing.
- Sampling is generally done at a fixed frequency, called the sampling frequency.**

Sampling Frequency Example



Ideal Sampling Frequency

- The ideal sampling frequency depends on the maximum frequency of the signal being digitized. If the sampling frequency is too low, then rapid changes in the analog signal may not be represented in the resulting digital data



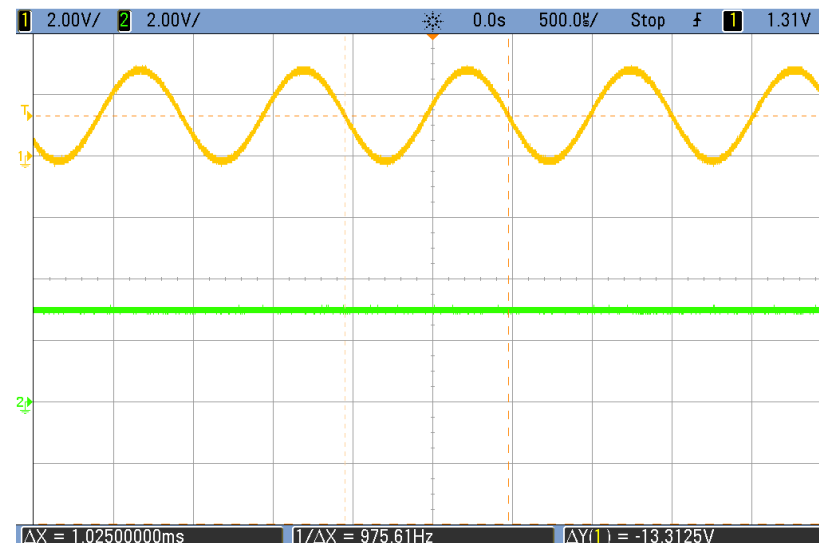
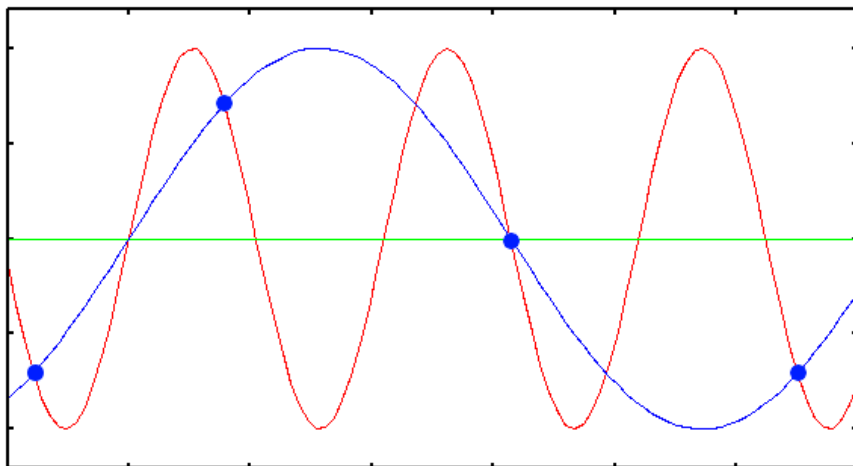
- Waveforms showing that undersampling results in the digitized output (lower traces) not representing the input waveforms (upper traces).
- This is known as Aliasing**

Nyquist-Shannon Sampling Theorem

- The Nyquist-Shannon sampling criterion states that the sampling frequency must be at least double that of the highest frequency of interest
 - **However, it is generally agreed that the sampling frequency should be 2 - 5 times the highest frequency found in the analog signal**
 - The name of the criterion is usually shortened to one or the other names – Nyquist criterion or Shannon Criterion.

Aliasing

- If the Nyquist criteria is not satisfied, a phenomenon known as **aliasing** occurs. Many weird and wonderful things can happen.
- Two signals can have the same digitized output.
- If the sampling frequency is the same as the input frequency, the output is constant!



Types of ADC

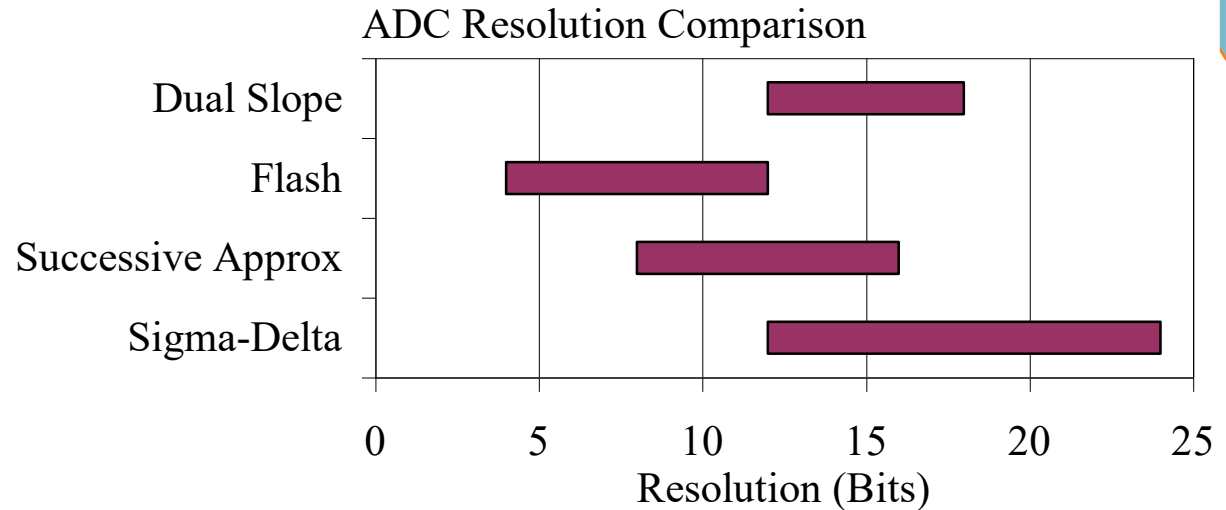
**Many different types
of ADC**

**All have different
attributes:**

- Resolution
- Speed
- Cost

Most Common:

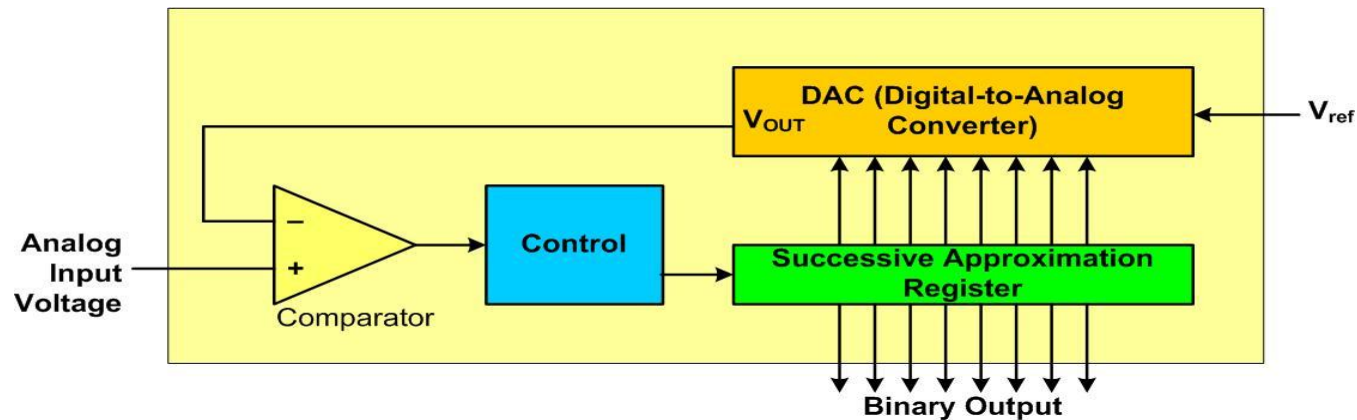
- Successive Approximation



Type	Speed (relative)	Cost (relative)
Dual Slope	Slow	Med
Flash	Very Fast	High
Successive Approx	Medium – Fast	Low
Sigma-Delta	Slow	Low

Successive Approximation ADC

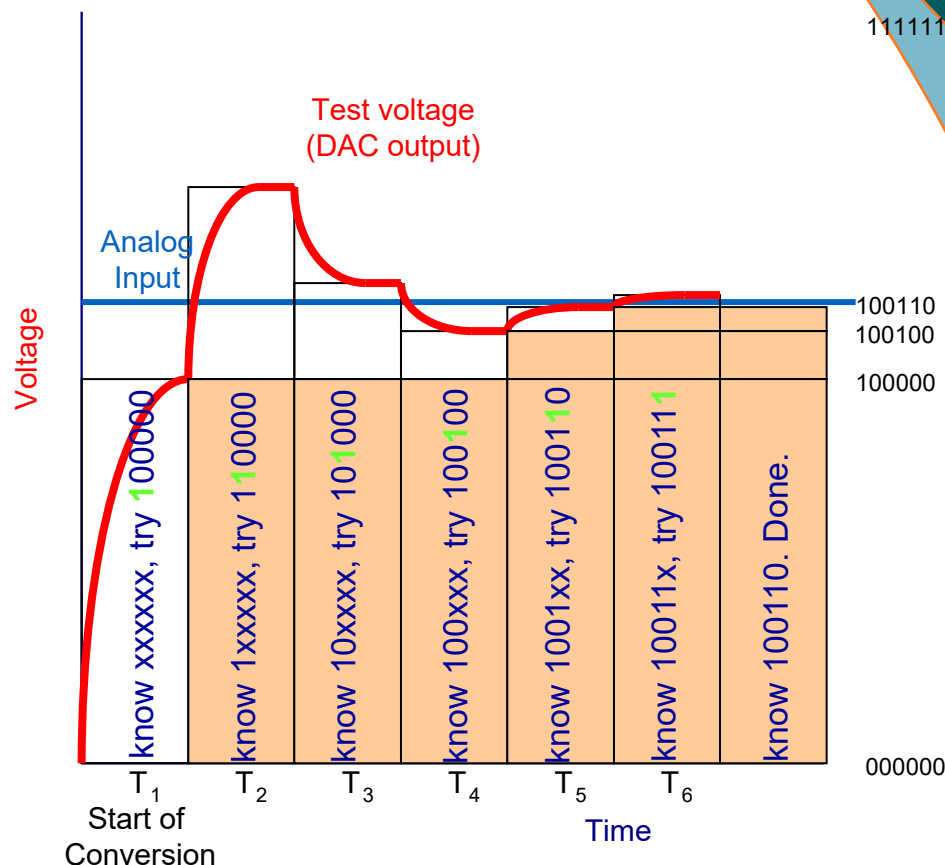
- Most popular type of ADC



- A Successive Approximation Register (SAR) is added to the circuit
- Instead of counting up in binary sequence, this register counts by trying all values of bits starting with the MSB and finishing at the LSB.
- The register monitors the comparators output to see if the binary count is greater or less than the analog signal input and adjusts the bits accordingly

ADC - Successive Approximation Conversion

- Successively approximate input voltage by using a binary search and a DAC
- SA Register holds current approximation of result
- At the beginning All DAC bits = 0
- Start with DAC's most significant bit = 1
 - If the DAC output (test voltage) is **smaller** than the input then set the current bit to 1, else clear the current bit to 0
- Repeat
 - Set next input bit for DAC to 1
 - Wait for DAC and comparator to stabilize
 - If the DAC output (test voltage) is **smaller** than the input then set the current bit to 1, else clear the current bit to 0



Successive Approximation

Advantages / Disadvantages

Advantages

- Capable of high speed and reliable
- Medium accuracy compared to other ADC types
- Good tradeoff between speed and cost
- Conversion time is fixed

Disadvantages

- Higher resolution successive approximation ADC's will be slower
- Speed limited to ~5Msps

Exercises

Exercises

A 10-bit ADC has a reference voltage of 3.3V ($V_{+ref} = 3.3V$, $V_{-ref} = 0V$)

Q1

What is the digital output for each of the following input voltages?

(i) $V_i = 3.0 V$

(ii) $V_i = 1.5V$

Q2

What is the analogue input voltage if the observed digital output is;

(i) 1000 decimal

(ii) 1FF Hex

Q3

What is the smallest change in input voltage, guaranteed to cause a change in the digital output (i.e what is the Quantization size (also called the resolution) of the ADC)?

MSP-EXP430G2ET LaunchPad ADC

MSP430 ADC Module

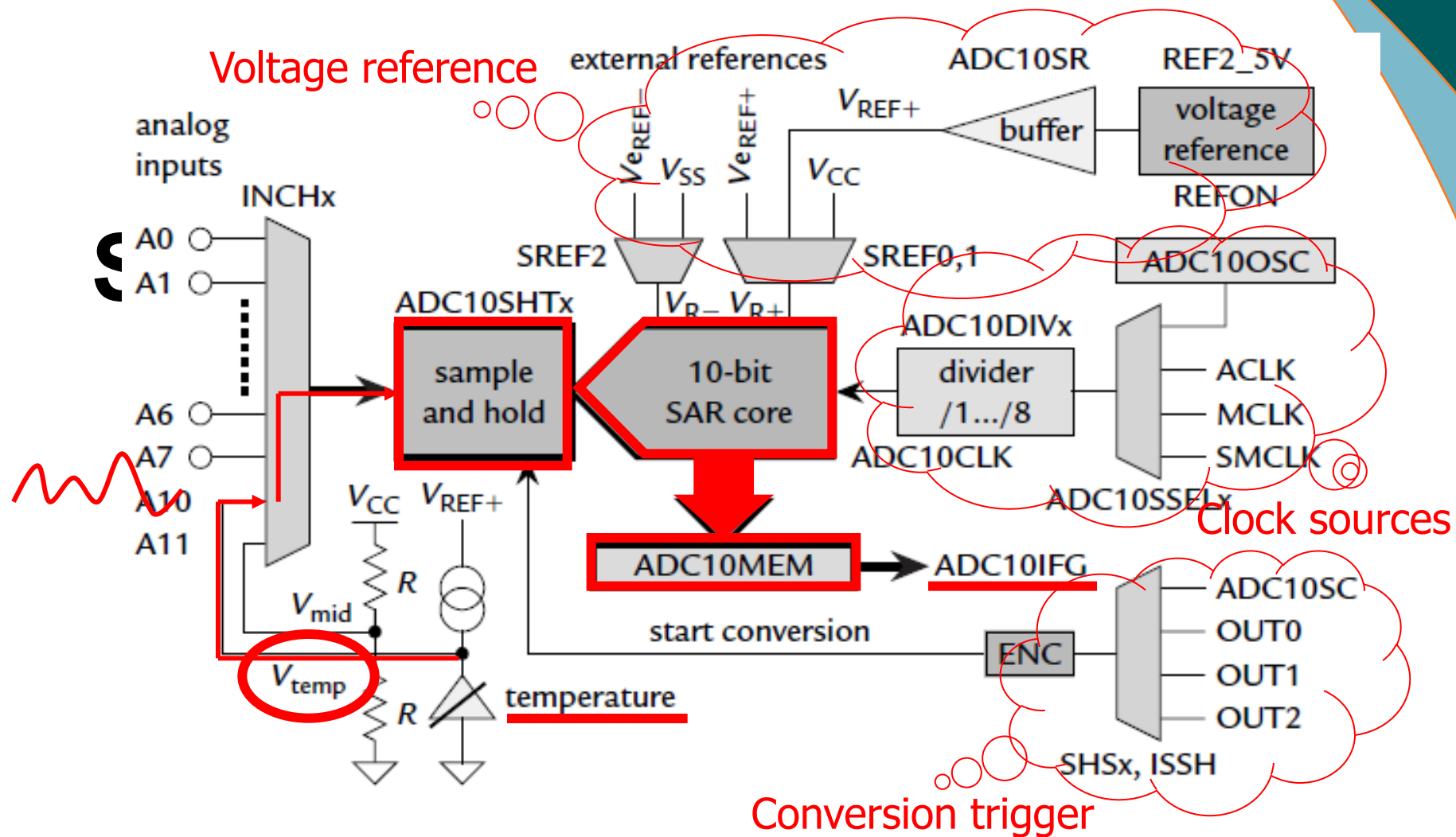
ADC10

- Mixed Signal Processing is a mixture of Analog signals/systems with digital (microcontroller) circuitry. On the MSP430-G2553 there are two “mixed signal” facilities;
 1. Analogue to Digital conversion on the MSP430-G2553
 - (ADC10 – 10-bit internal Analogue to Digital Converter)
 2. Analog Comparator implemented on every MSP430 series.
- We will look at the most useful of these – the Analogue to Digital Converter- ADC10

Notes on MSP430 ADC

- The ADC (ADC10) described in the following slides is implemented on
 - (i) the MSP430_G2553 (Both versions of LaunchPad)
 - (ii) the MSP430_G2452
- Other versions of the MSP430 may implement different ADCs – ADC12 or SD-16, SD-16A, etc, etc.
- Some basic MSP430s do not incorporate any ADC .
- CHECK USER MANUAL of your own device for details.

MSP430 ADC Layout



MSP430 ADC

- We will learn about the various registers associated with the MSP430 and how to program them in the Lab.

MSP430 ADC Registers

ADC 10 Register Table (p598 MSP430G2xx User Manual)

22.3 ADC10 Registers

Table 22-3 lists the memory-mapped registers for the ADC10.

Table 22-3. ADC10 Registers

Address	Acronym	Register Name	Type	Reset	Section
1B0h	ADC10CTL0	ADC10 control 0	Read/write	00h with POR	Section 22.3.1
1B2h	ADC10CTL1	ADC10 control 1	Read/write	00h with POR	Section 22.3.2
4Ah	ADC10AE0	ADC10 input enable 0	Read/write	00h with POR	Section 22.3.3
4Bh	ADC10AE1	ADC10 input enable 1	Read/write	00h with POR	Section 22.3.4
1B4h	ADC10MEM	ADC10 memory	Read	Unchanged	Section 22.3.5
48h	ADC10DTC0	ADC10 data transfer control 0	Read/write	00h with POR	Section 22.3.6
49h	ADC10DTC1	ADC10 data transfer control 1	Read/write	00h with POR	Section 22.3.7
1BCh	ADC10SA	ADC10 data transfer start address	Read/write	200h with POR	Section 22.3.8

MSP430 ADC Registers

ADC10CTL0 (p599 MSP430G2xx User Manual)

Figure 22-17. ADC10CTL0 Register

15	14	13	12	11	10	9	8
SREFx			ADC10SHTx		ADC10SR	REFOUT	REFBURST
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MSC	REF2_5V	REFON	ADC10ON	ADC10IE	ADC10IFG	ENC	ADC10SC
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
15-13	SREFx	R/W	0h	Select reference. Can be modified only when ENC = 0. 000b = $V_{R+} = V_{CC}$ and $V_{R-} = V_{SS}$ 001b = $V_{R+} = V_{REF+}$ and $V_{R-} = V_{SS}$ 010b = $V_{R+} = V_{eREF+}$ and $V_{R-} = V_{SS}$. Devices with V_{eREF+} only. 011b = $V_{R+} = \text{Buffered } V_{eREF+}$ and $V_{R-} = V_{SS}$. Devices with V_{eREF+} pin only. 100b = $V_{R+} = V_{CC}$ and $V_{R-} = V_{REF-} / V_{eREF-}$. Devices with V_{eREF-} pin only. 101b = $V_{R+} = V_{REF+}$ and $V_{R-} = V_{REF-} / V_{eREF-}$. Devices with V_{eREF+} and V_{eREF-} pins only. 110b = $V_{R+} = V_{eREF+}$ and $V_{R-} = V_{REF-} / V_{eREF-}$. Devices with V_{eREF+} and V_{eREF-} pins only. 111b = $V_{R+} = \text{Buffered } V_{eREF+}$ and $V_{R-} = V_{REF-} / V_{eREF-}$. Devices with V_{eREF+} and V_{eREF-} pins only.

MSP430 ADC Registers

ADC10CTL0 (p599 MSP430G2xx User Manual)

Figure 22-17. ADC10CTL0 Register

15	14	13	12	11	10	9	8
SREFx			ADC10SHTx		ADC10SR	REFOUT	REFBURST
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MSC	REF2_5V	REFON	ADC10ON	ADC10IE	ADC10IFG	ENC	ADC10SC
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
12-11	ADC10SHTx	R/W	0h	ADC10 sample-and-hold time. Can be modified only when ENC = 0. 00b = 4 ADC10CLK cycles 01b = 8 ADC10CLK cycles 10b = 16 ADC10CLK cycles 11b = 64 ADC10CLK cycles
10	ADC10SR	R/W	0h	ADC10 sampling rate. This bit selects the reference buffer drive capability for the maximum sampling rate. Setting ADC10SR reduces the current consumption of the reference buffer. Can be modified only when ENC = 0. 0b = Reference buffer supports up to approximately 200 ksp/s 1b = Reference buffer supports up to approximately 50 ksp/s
9	REFOUT	R/W	0h	Reference output. Can be modified only when ENC = 0. 0b = Reference output off 1b = Reference output on. Devices with V_{eREF+} / V_{REF+} pin only.

MSP430 ADC Registers

ADC10CTL0 (p599 MSP430G2xx User Manual)

Figure 22-17. ADC10CTL0 Register

15	14	13	12	11	10	9	8
SREFx			ADC10SHTx		ADC10SR	REFOUT	REFBURST
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MSC	REF2_5V	REFON	ADC10ON	ADC10IE	ADC10IFG	ENC	ADC10SC
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
8	REFBURST	R/W	0h	Reference burst. Can be modified only when ENC = 0. 0b = Reference buffer on continuously 1b = Reference buffer on only during sample-and-conversion
7	MSC	R/W	0h	Multiple sample and conversion. Valid only for sequence or repeated modes. Can be modified only when ENC = 0. 0b = The sampling requires a rising edge of the SHI signal to trigger each sample-and-conversion. 1b = The first rising edge of the SHI signal triggers the sampling timer, but further sample-and-conversions are performed automatically as soon as the prior conversion is completed
6	REF2_5V	R/W	0h	Reference-generator voltage. REFON must also be set. Can be modified only when ENC = 0. 0b = 1.5 V 1b = 2.5 V

MSP430 ADC Registers

ADC10CTL0 (p599 MSP430G2xx User Manual)

Figure 22-17. ADC10CTL0 Register

15	14	13	12	11	10	9	8
SREFx			ADC10SHTx		ADC10SR	REFOUT	REFBURST
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MSC	REF2_5V	REFON	ADC10ON	ADC10IE	ADC10IFG	ENC	ADC10SC
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
5	REFON	R/W	0h	Reference generator on. Can be modified only when ENC = 0. 0b = Reference off 1b = Reference on
4	ADC10ON	R/W	0h	ADC10 on. Can be modified only when ENC = 0. 0b = ADC10 off 1b = ADC10 on
3	ADC10IE	R/W	0h	ADC10 interrupt enable 0b = Interrupt disabled 1b = Interrupt enabled

MSP430 ADC Registers

ADC10CTL0 (p599 MSP430G2xx User Manual)

Figure 22-17. ADC10CTL0 Register

15	14	13	12	11	10	9	8
SREFx			ADC10SHTx		ADC10SR	REFOUT	REFBURST
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
MSC	REF2_5V	REFON	ADC10ON	ADC10IE	ADC10IFG	ENC	ADC10SC
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
2	ADC10IFG	R/W	0h	ADC10 interrupt flag. This bit is set if ADC10MEM is loaded with a conversion result. It is automatically reset when the interrupt request is accepted, or it may be reset by software. When using the DTC, this flag is set when a block of transfers is completed. 0b = No interrupt pending 1b = Interrupt pending
1	ENC	R/W	0h	Enable conversion 0b = ADC10 disabled 1b = ADC10 enabled
0	ADC10SC	R/W	0h	Start conversion. Software-controlled sample-and-conversion start. ADC10SC and ENC may be set together with one instruction. ADC10SC is reset automatically. 0b = No sample-and-conversion start 1b = Start sample-and-conversion

MSP430 ADC Registers

ADC10CTL1(p601 MSP430G2xx User Manual)

Figure 22-18. ADC10CTL1 Register

15	14	13	12	11	10	9	8
INCHx				SHSx		ADC10DF	ISSH
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
ADC10DIVx			ADC10SSELx		CONSEQx		ADC10BUSY
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	r-0
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
15-12	INCHx	R/W	0h	Input channel select. These bits select the channel for a single-conversion or the highest channel for a sequence of conversions. Only available ADC channels should be selected. See the device-specific data sheet. Can be modified only when ENC = 0. 0000b = A0 0001b = A1 0010b = A2 0011b = A3 0100b = A4 0101b = A5 0110b = A6 0111b = A7 1000b = V_{eREF+} 1001b = V_{REF-}/V_{eREF-} 1010b = Temperature sensor 1011b = $(V_{CC} - V_{SS}) / 2$ 1100b = $(V_{CC} - V_{SS}) / 2$, A12 on MSP430F22xx, MSP430G2x44, and MSP430G2x55 devices 1101b = $(V_{CC} - V_{SS}) / 2$, A13 on MSP430F22xx, MSP430G2x44, and MSP430G2x55 devices 1110b = $(V_{CC} - V_{SS}) / 2$, A14 on MSP430F22xx, MSP430G2x44, and MSP430G2x55 devices 1111b = $(V_{CC} - V_{SS}) / 2$, A15 on MSP430F22xx, MSP430G2x44, and MSP430G2x55 devices

MSP430 ADC Registers

ADC10CTL1(p601 MSP430G2xx User Manual)

Figure 22-18. ADC10CTL1 Register

15	14	13	12	11	10	9	8
INCHx				SHSx		ADC10DF	ISSH
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
ADC10DIVx			ADC10SSELx		CONSEQx		ADC10BUSY
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	r-0
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
11-10	SHSx	R/W	0h	Sample-and-hold source select. Can be modified only when ENC = 0. 00b = ADC10SC bit 01b = Timer_A.OUT1 ⁽¹⁾ 10b = Timer_A.OUT0 ⁽¹⁾ 11b = Timer_A.OUT2 (Timer_A.OUT1 on MSP430F20x0, MSP430G2x31, and MSP430G2x30 devices) ⁽¹⁾
9	ADC10DF	R/W	0h	ADC10 data format. Can be modified only when ENC = 0. 0b = Straight binary 1b = 2s complement
8	ISSH	R/W	0h	Invert signal sample-and-hold. Can be modified only when ENC = 0. 0b = The sample-input signal is not inverted. 1b = The sample-input signal is inverted.

MSP430 ADC Registers

ADC10CTL1(p601 MSP430G2xx User Manual)

Figure 22-18. ADC10CTL1 Register

15	14	13	12	11	10	9	8
INCHx				SHSx		ADC10DF	ISSH
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
ADC10DIVx			ADC10SSELx		CONSEQx		ADC10BUSY
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	r-0
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
7-5	ADC10DIVx	R/W	0h	ADC10 clock divider. Can be modified only when ENC = 0. 000b = /1 001b = /2 010b = /3 011b = /4 100b = /5 101b = /6 110b = /7 111b = /8
4-3	ADC10SSELx	R/W	0h	ADC10 clock source select. Can be modified only when ENC = 0. 00b = ADC10OSC 01b = ACLK 10b = MCLK 11b = SMCLK

MSP430 ADC Registers

ADC10CTL1(p601 MSP430G2xx User Manual)

Figure 22-18. ADC10CTL1 Register

15	14	13	12	11	10	9	8
INCHx				SHSx		ADC10DF	ISSH
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)
7	6	5	4	3	2	1	0
ADC10DIVx			ADC10SSELx		CONSEQx		ADC10BUSY
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	r-0
Can be modified only when ENC = 0							

Bit	Field	Type	Reset	Description
2-1	CONSEQx	R/W	0h	Conversion sequence mode select 00b = Single-channel single-conversion mode 01b = Sequence-of-channels mode 10b = Repeat-single-channel mode 11b = Repeat-sequence-of-channels mode
0	ADC10BUSY	R	0h	ADC10 busy. This bit indicates an active sample or conversion operation 0b = No operation is active. 1b = A sequence, sample, or conversion is active.

MSP430 ADC Registers

ADC10AE0 (p603 MSP430G2xx User Manual)

Figure 22-19. ADC10AE0 Register

7	6	5	4	3	2	1	0
ADC10AE0x							
rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)	rw-(0)

Table 22-6. ADC10AE0 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	ADC10AE0x	R/W	0h	ADC10 analog enable. These bits enable the corresponding pin for analog input. Bit 0 corresponds to A0, Bit 1 corresponds to A1, and so on. The analog enable bit of not implemented channels should not be programmed to 1. 0b = Analog input disabled 1b = Analog input enabled

MSP430 ADC Registers

ADC10MEM (p604 MSP430Gxx User Manual)

Conversion-Memory Register. This register is read as either right-justified straight-binary format or left-justified 2s-complement format, depending on the value of the ADC10DF bit in the ADC10CTL1 register.

Figure 22-21. ADC10MEM Register

15	14	13	12	11	10	9	8
Conversion_Results							
r0	r0	r0	r0	r0	r0	r	r
7	6	5	4	3	2	1	0
Conversion_Results							
r	r	r	r	r	r	r	r

Table 22-8. ADC10MEM Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	Conversion_Results	R	Unchanged	If ADC10DF = 0, the 10-bit conversion results are right-justified straight-binary format. Bit 9 is the MSB. 15-10 are always 0. If ADC10DF = 1, the 10-bit conversion results are left-justified 2s-complement format. Bit 15 is the MSB. Bits 5-0 are always 0.

